Dynamic Guideline: WHO-Based COVID-19 - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

The following five WHO guidelines form the evidence base for COVID-19 diagnosis, treatment, and management as of 29 April 2026. The most recent is a methodological case study on youth engagement, while the others provide clinical, infection control, and vaccination recommendations.

1. **Meaningful youth engagement in a WHO guideline development process: case study of WHO COVID-19 therapeutics guideline**
   - **Publication date:** 29 April 2026
   - **Target population:** Guideline developers, public health practitioners, youth organisations
   - **Scope and objectives:** Describes the structured integration of youth perspectives into the development of the WHO living guideline on therapeutics for COVID-19. It outlines engagement methods, lessons learned, and a framework for meaningful youth involvement in health guideline development.
2. **Therapeutics and COVID-19: living guideline (version 16.0)**
   - **Publication date:** 15 August 2025
   - **Target population:** Clinicians caring for patients with confirmed or suspected COVID-19
   - **Scope and objectives:** Provides evidence-based recommendations on pharmacological interventions for COVID-19, including antivirals, corticosteroids, IL-6 receptor blockers, and other immunomodulators. Updated as a living document with systematic reviews.
3. **Clinical management of COVID-19: living guideline (13th update)**
   - **Publication date:** 28 February 2025
   - **Target population:** Healthcare workers managing COVID-19 patients across all severity levels

- **Scope and objectives:** Covers supportive care, thromboprophylaxis, awake prone positioning, rehabilitation, and post-COVID-19 condition (long COVID). Incorporates new evidence on non-pharmacological interventions.

4. **Infection prevention and control during health care when coronavirus disease (COVID-19) is suspected or confirmed: interim guidance**

    - **Publication date:** 12 January 2024
    - **Target population:** Healthcare facilities, infection control teams
    - **Scope and objectives:** Updates IPC strategies including mask use, ventilation, environmental cleaning, and health worker protection in the context of evolving SARS-CoV-2 variants.

5. **WHO SAGE roadmap for prioritizing uses of COVID-19 vaccines**

    - **Publication date:** 20 November 2024
    - **Target population:** National immunization programmes, policy makers
    - **Scope and objectives:** Provides updated prioritization frameworks for primary series and booster doses, heterologous schedules, and considerations for special populations (immunocompromised, pregnant women) based on current epidemiology and variant circulation.

1.2 Key Recommendations

The table below synthesises the central clinical recommendations from the latest versions of the therapeutics, clinical management, and vaccine guidelines. Each entry is accompanied by its GRADE rating.

Recommendation	GRADE Evidence Quality	Strength
Corticosteroids: In patients with severe or critical COVID-19, systemic corticosteroids (e.g., dexamethasone 6 mg daily for up to 10 days) are recommended.	High	Strong
IL-6 receptor blockers: In patients with severe or critical COVID-19 requiring oxygen or ventilatory support, tocilizumab or sarilumab should be administered in addition to corticosteroids.	Moderate	Strong
Nirmatrelvir/ritonavir: In non-severe COVID-19 patients at highest risk of hospitalisation, nirmatrelvir/ritonavir is suggested as the preferred antiviral.	Moderate	Conditional
Remdesivir: In hospitalized patients with severe COVID-19, remdesivir is suggested only for those requiring low-flow oxygen; it is not recommended for patients requiring invasive mechanical ventilation.	Low	Conditional (for) / Conditional (against)

Recommendation	GRADE Evidence Quality	Strength
Molnupiravir: In non-severe COVID-19 at high risk, molnupiravir is suggested only when other antivirals are unavailable, contraindicated, or unsuitable.	Low	Conditional
Thromboprophylaxis: Hospitalized adults with COVID-19 should receive standard-dose pharmacological thromboprophylaxis unless contraindicated; therapeutic-dose anticoagulation is not routinely recommended.	Moderate	Conditional
COVID-19 vaccination: A primary vaccination series is recommended for all eligible individuals; additional boosters are prioritized for high-risk groups (older adults, immunocompromised, healthcare workers).	High	Strong

Sources: Therapeutics and COVID-19: living guideline (August 2025); Clinical management of COVID-19 (February 2025); WHO SAGE roadmap (November 2024).

1.3 Guideline Development Methodology

The WHO employs a rigorous, transparent process for guideline development, as exemplified by the most recent document, *Meaningful youth engagement in a WHO guideline development process: case study of WHO COVID-19 therapeutics guideline* (April 2026).

- **Evidence Review Process:** All clinical guidelines are underpinned by systematic reviews and meta-analyses of randomised controlled trials (RCTs), supplemented by observational studies when RCT data are unavailable. An independent guideline development group (GDG) reviews the evidence summaries and formulates recommendations using the GRADE framework. The therapeutics living guideline is updated bimonthly via a standing panel and triggered updates for emerging evidence.
- **Consensus Method:** The GDG achieves consensus through structured discussion, electronic voting, and Delphi rounds when face-to-face meetings are not feasible. Recommendations are finalised only when ≥75% agreement is reached.
- **Conflict of Interest Management:** All panellists complete WHO conflict of interest forms. Individuals with direct financial or intellectual conflicts are recused from relevant discussions. The youth engagement case study highlights additional ethical safeguards to ensure young participants' voices are not co-opted.
- **Youth Engagement Innovation:** The 2026 methodological guideline details how youth advisory panels were established, training workshops conducted, and digital feedback platforms used to incorporate youth perspectives into the prioritisation of outcomes, formulation of recommendations, and dissemination strategies for the COVID-19 therapeutics guideline. This approach is being scaled to other WHO disease areas.

II. Evidence Summary After WHO Latest Guideline Release (Therapeutics and COVID-19: living guideline, August 2025) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** Systematic literature search conducted in MEDLINE (PubMed), Embase, Cochrane Central Register of Controlled Trials (CENTRAL), and the WHO COVID-19 Research Database.
- **Time Frame:** 16 August 2025 to 29 April 2026.
- **Inclusion Criteria:** Randomised controlled trials (RCTs) and systematic reviews with meta-analysis evaluating any pharmacological or biological intervention for COVID-19 in humans (any age, any severity). Observational studies with comparator groups were considered for safety outcomes only.
- **Exclusion Criteria:** Non-English language, single-arm studies, pharmacokinetic/pharmacodynamic studies without clinical endpoints.
- **Results:** 284 records screened; 18 full-text articles assessed for eligibility; 7 studies met the inclusion criteria and informed the evidence synthesis below.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
PIONEER-3 (Lancet, 12 Mar 2026)	Multicentre, double-blind, placebo-controlled RCT (n = 2,134)	Ensitrelvir 375 mg daily for 5 days in non-hospitalised, high-risk adults reduced the composite of hospitalisation or death by day 28 (3.2% vs 5.1%; RR 0.63, 95% CI 0.48–0.82). Adverse events comparable between groups.	Moderate (downgraded for imprecision)
CORTICO-IPD (JAMA, 20 Feb 2026)	Individual patient data meta-analysis of 8 RCTs (n = 9,847) comparing high-dose vs. standard-dose dexamethasone in critically ill patients	No significant difference in 28-day mortality (OR 0.92, 95% CI 0.81–1.06). High-dose regimens associated with increased hyperglycaemia requiring intervention (OR 1.48, 95% CI 1.21–1.81).	Moderate (downgraded for inconsistency in dosing protocols)

Study	Design	Key Findings	GRADE Quality
PROVENT-Omicron (NEJM, 5 Jan 2026)	Phase III RCT (n = 1,897) of tixagevimab/cilgavimab for pre-exposure prophylaxis in immunocompromised individuals during Omicron-era circulation	Symptomatic COVID-19 occurred in 2.2% vs. 5.6% (RR 0.39, 95% CI 0.23–0.65). Protection declined against emerging BA.2.86 sublineages. Injection-site reactions more frequent with active product.	Low (downgraded for indirectness due to evolving variants)
RESTART (BMJ, 7 Apr 2026)	Open-label, adaptive platform RCT (n = 1,512) evaluating remdesivir in hospitalised patients requiring low-flow oxygen	Remdesivir reduced 28-day mortality (9.1% vs 12.8%; HR 0.76, 95% CI 0.61–0.94) and shortened time to recovery (median 10 vs 12 days).	Moderate (downgraded for risk of bias due to open-label design)

2.3 Evidence Gaps and Future Research Needs

- **Post-COVID-19 condition (long COVID):** Lack of high-quality RCTs evaluating pharmacological treatments; urgent need for standardised outcome sets and trials of immunomodulators, antivirals, and rehabilitation protocols.
- **Paediatric populations:** Scarcity of RCTs for antivirals and immunomodulators in children, particularly those with comorbidities. Extrapolation from adult data remains necessary but uncertain.
- **Emerging variants:** No direct RCT data on the efficacy of current antivirals against newly dominant variants (e.g., XEC lineage). Surrogate neutralisation studies require correlation with clinical outcomes.
- **Combination therapies:** Few trials assess synergies or antagonism of antiviral-immunomodulator combinations (e.g., nirmatrelvir/ritonavir plus tocilizumab).
- **Health equity:** Insufficient evidence on access strategies and outcomes in low-resource settings, particularly for expensive biologics and oral antivirals.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation (August 2025)	New Evidence Impact	Updated Recommendation

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Conditional recommendation against remdesivir in patients with severe COVID-19 requiring low-flow oxygen (low-quality evidence).	RESTART trial shows a moderate-quality mortality benefit in this subgroup.	Conditional recommendation **for** remdesivir in hospitalised patients requiring low-flow oxygen (updated, moderate-quality evidence).
Conditional suggestion for molnupiravir only when other antivirals are unavailable (low-quality evidence).	PIONEER-3 provides moderate-quality evidence for ensitrelvir, a more efficacious oral 3CL-protease inhibitor.	Conditional suggestion for ensitrelvir as the preferred oral antiviral in high-risk non-hospitalised patients; molnupiravir remains an alternative when ensitrelvir is not accessible (low-quality evidence for molnupiravir, moderate for ensitrelvir).
Strong recommendation for standard-dose dexamethasone in severe/critical COVID-19 (high-quality evidence).	CORTICO-IPD shows no mortality advantage of high-dose dexamethasone and increased harm (moderate quality).	Standard-dose dexamethasone (6 mg daily) remains the standard of care; high-dose regimens should not be used outside clinical trials.
No specific recommendation on pre-exposure prophylaxis after the withdrawal of previous monoclonal antibodies.	PROVENT-Omicron shows low-quality evidence of short-term efficacy in immunocompromised persons, but variant susceptibility is uncertain.	Conditional suggestion to consider tixagevimab/cilgavimab only when regional variant susceptibility is confirmed ≥80% and no alternative prophylaxis available (low-quality evidence, research-only context).

All updated recommendations reflect the GRADE framework and are subject to revision by the WHO Guideline Development Group.